

Data-driven weather models: *A new era in meteorology*

Mariana Clare

on behalf of many colleagues across ECMWF

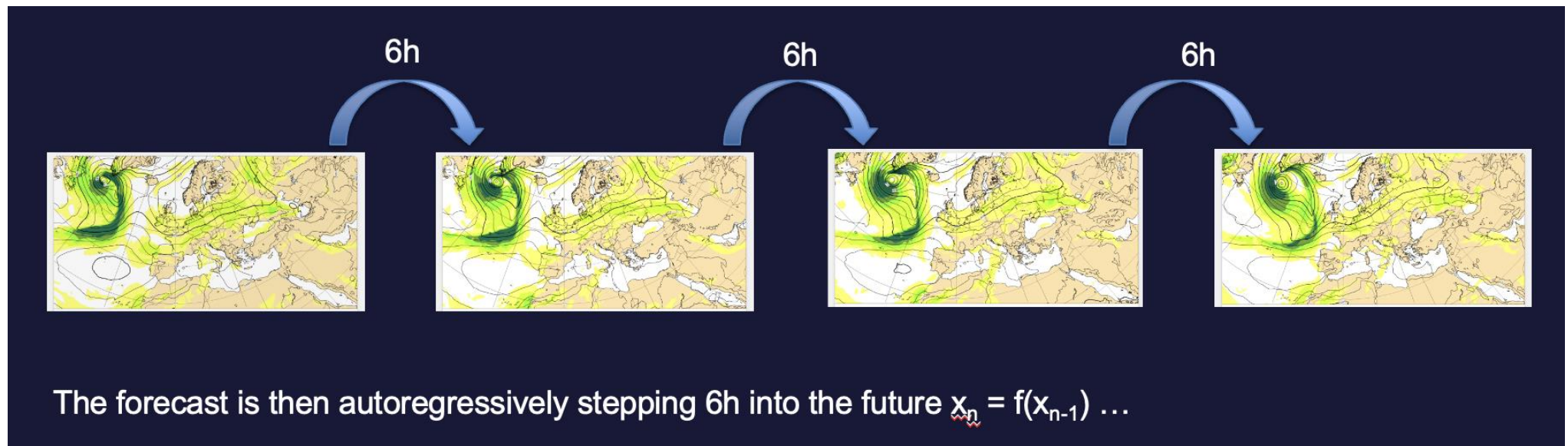
Classical NWP



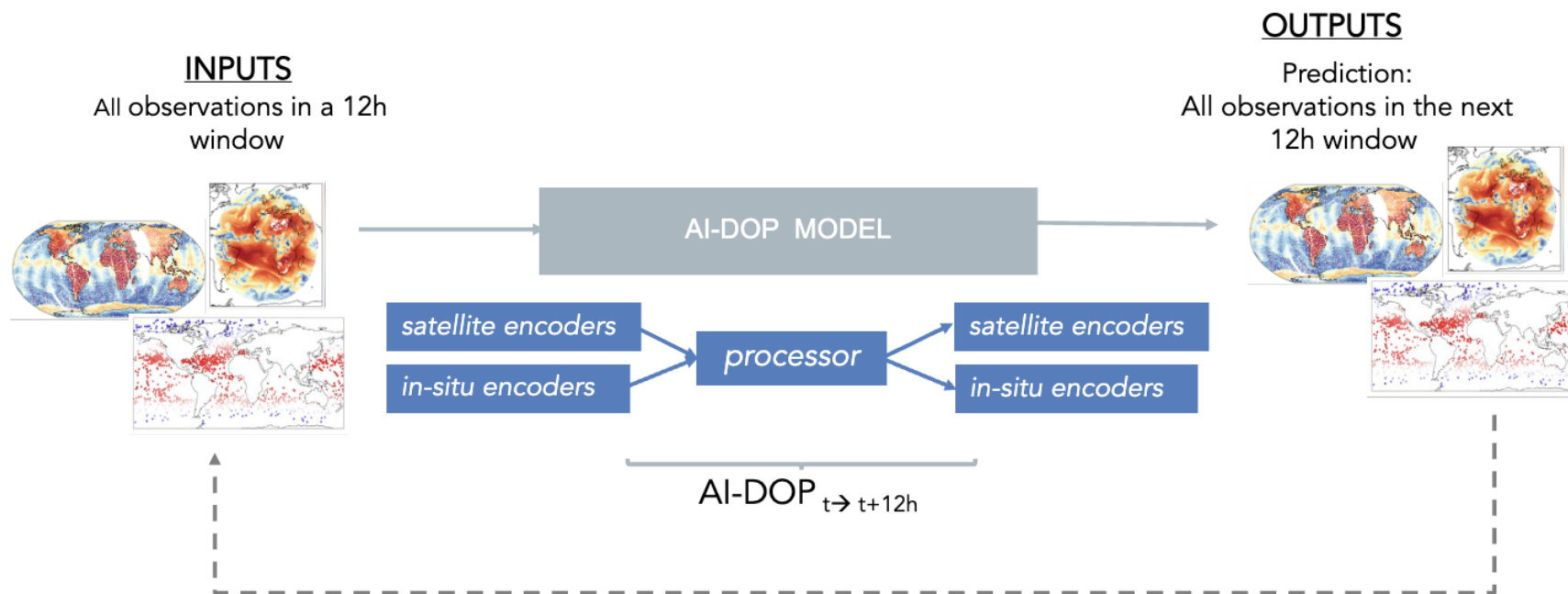
Data-driven weather forecasting system



Most data-driven weather forecasting models are trained on reanalysis datasets, specifically ERA5 produced as part of the Copernicus Climate Change Service (C3S) at ECMWF



Data-driven weather forecasting system



Training on observations (e.g. SYNOPs) at arbitrary locations allows for generalisation to standard gridded model output

A short history of data-driven weather forecasting

February 2022 – First competitive medium-range systems

- **Keisler** – GraphNN, competitive with GFS (USA)
- **NVIDIA** – **FourCastNet** Fourier+ , 0.25°, $O(10^4)$ faster & more energy efficient than IFS

December 2022

- **Deepmind** – **GraphCast**

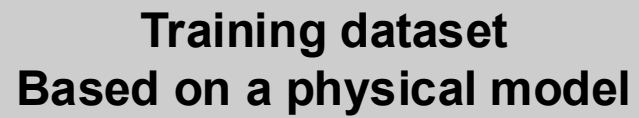
GraphNN
0.25° Many parameters with comparable skill to IFS.

November 2022

- **Huawei** – **PanguWeather**

Vision Transformer
0.25° “More accurate tropical cyclone tracks” than the IFS.

2018 – Concept explored (ECMWF and others)...



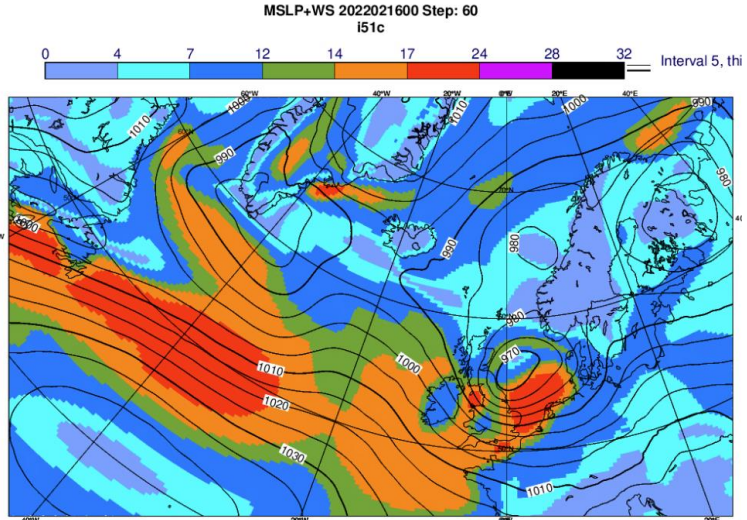
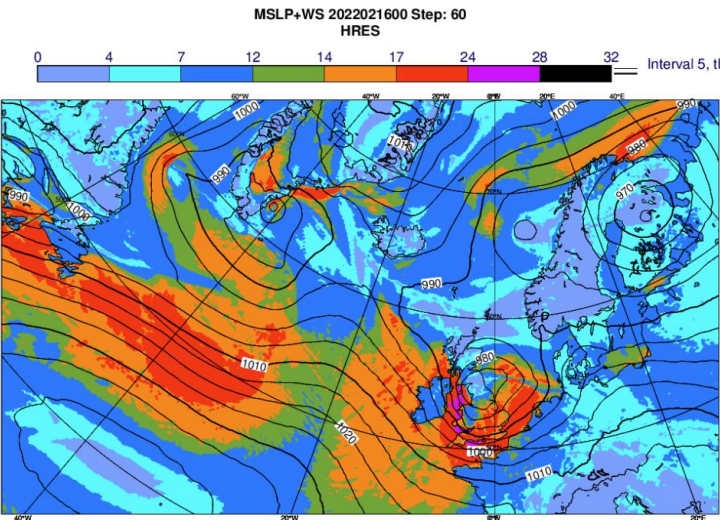
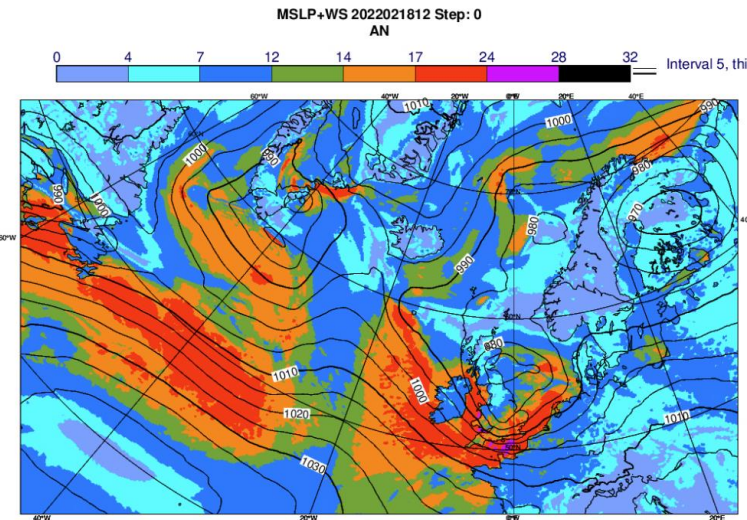
0.3
per forecast



Analysis

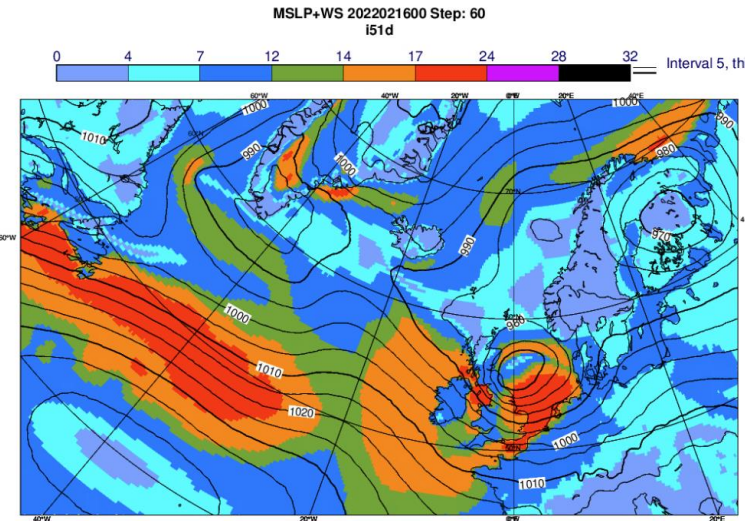
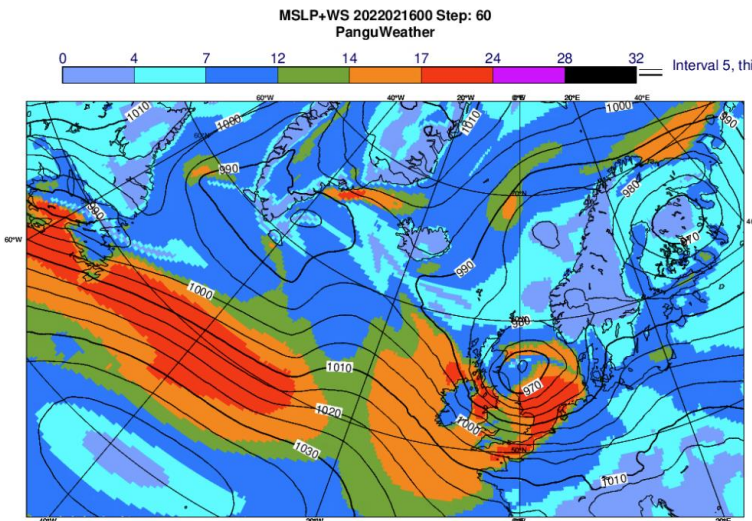
HRES

FourCastnet



PanguWeather

GraphCast



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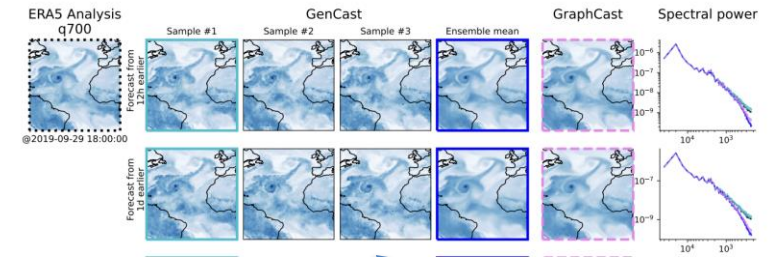
January-June 2023

- **Microsoft** – **ClimaX**
- **China academia/Shanghai Met** – **FengWu**
- **Alibaba** – **SwinRDM**
- **NVIDIA** – **SFNO**
- ...

December 2023

- **Deepmind** – **GenCast**

Probabilistic forecast (ensemble) – 1°
”Outperforming the leading operational ensemble forecast” (aka ECMWF)

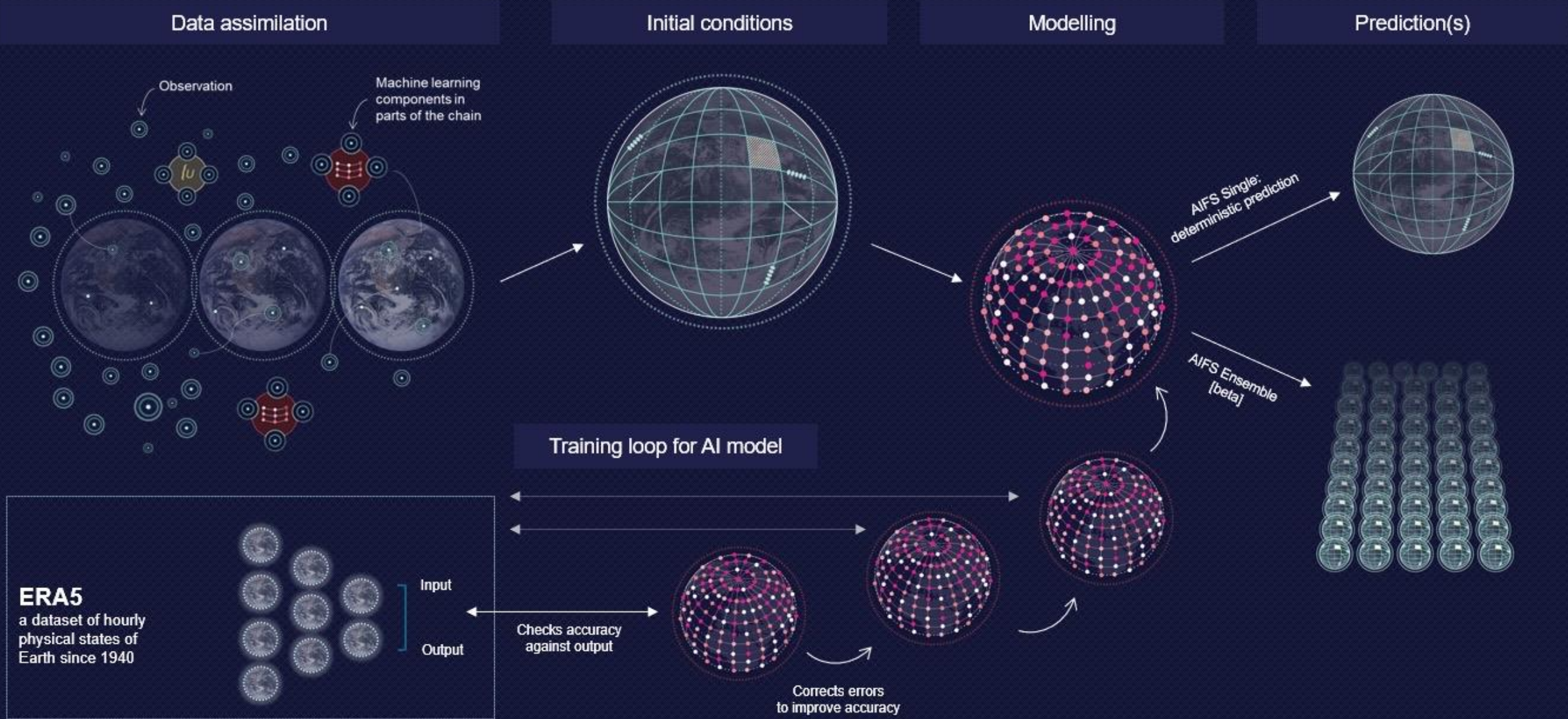


2018 – Concept explored (ECMWF and others)...

June 2023
ECMWF – ML project begins

Early 2023
Prototype AIFS developments begin

AIFS: Artificial Intelligence Forecasting System



A short history of data-driven weather forecasting

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December 2023

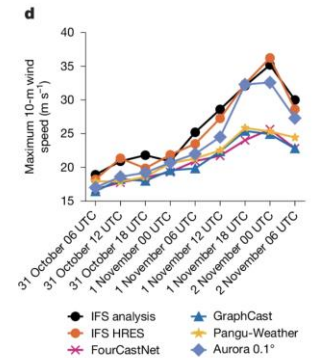
- **Deepmind** – GenCast

Probabilistic forecast (ensemble) – 0.25°
”Outperforming the leading operational ensemble forecast” (aka ECMWF)

June 2024

- **Microsoft** – Aurora

Higher resolution – 0.1°
Atmospheric composition



2018 – Concept explored (ECMWF and others)...

June 2023
ML project begins

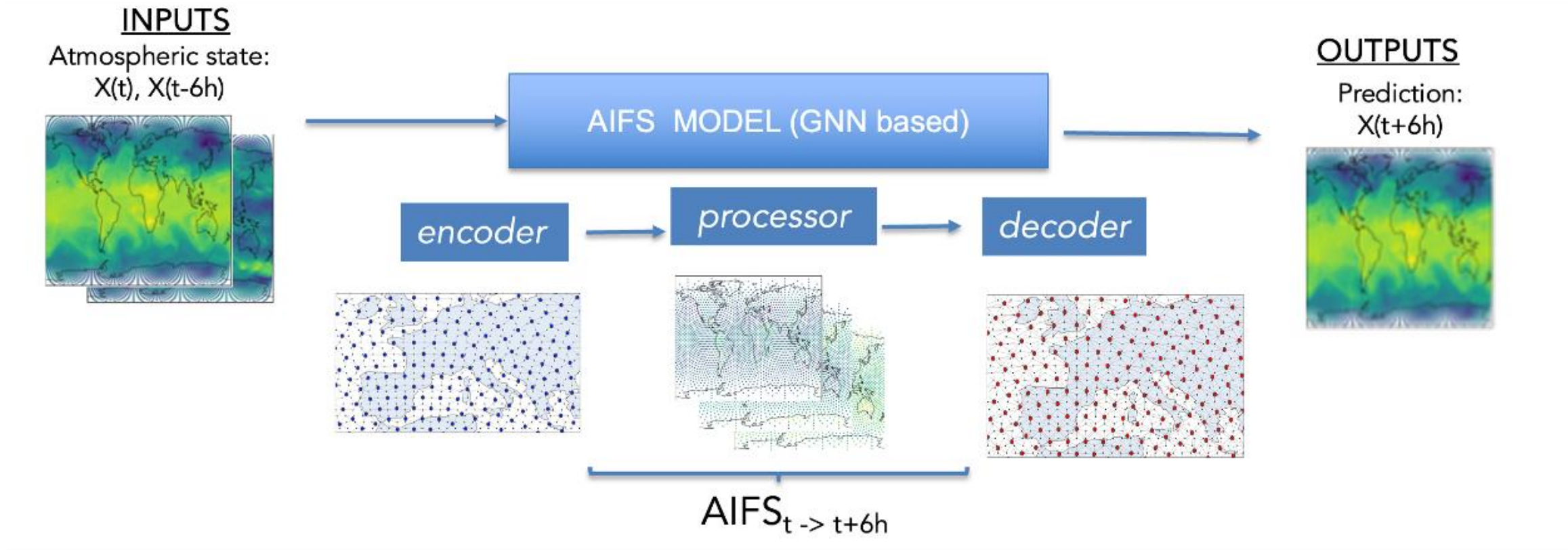
Jan/Feb 2024
AIFS first updates

July 2025:
AIFS ENS CRPS
1 operational

Early 2023
Prototype AIFS
developments
begin

October 2023
AIFS experimental
forecasts live

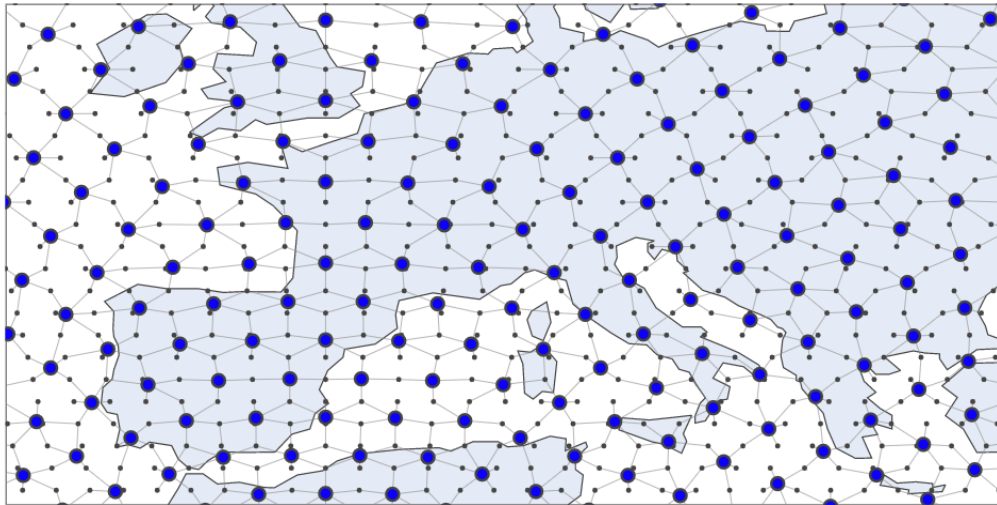
Feb 2025:
AIFS Single 1
operational



AIFS – Encoder and Decoder

AIFS works with the native IFS reduced gaussian grids ;
possible to split model and input data across multiple GPUs to
handle large memory requirements

Encoder, GNN



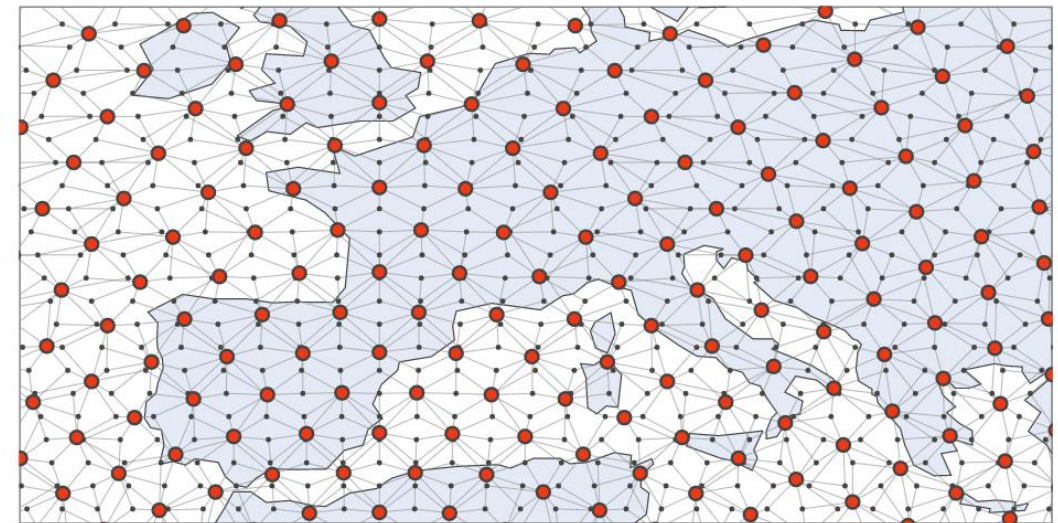
ERA5 – n320

~ 540 000 Nodes,
~ 1 Million Edges

16 Processor Layers

O96 ~ 40 000 Nodes

Decoder, GNN



ERA5 – n320

AIFS – Processor

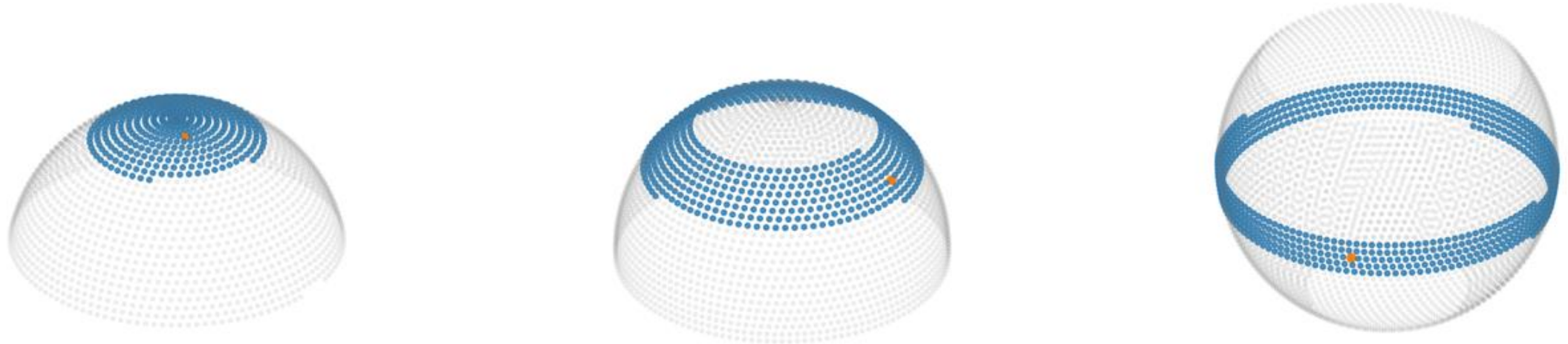
Transformer (like LLMs) that works with a sliding attention window -> attention bands around the globe

Red: target node

Blue: Nodes target node attends to in one processor layer

Grey: How far information can travel within e.g. 6 processor layers

(here lower resolution processor grid than operational AIFS for visualization)

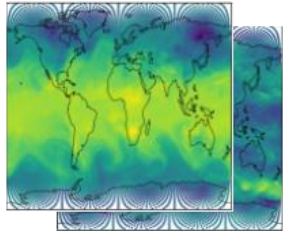


AIFS Single v1 - Training

TRAINING

INPUTS

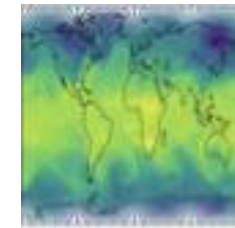
Atmospheric state:
 $X(t)$, $X(t-6h)$



AIFS MODEL (GNN based)

OUTPUTS

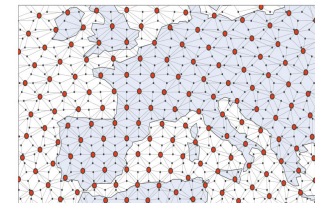
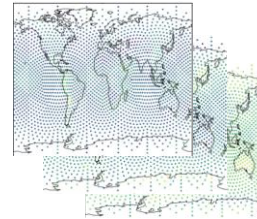
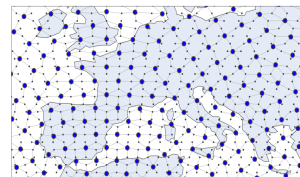
Prediction:
 $X(t+6h)$



encoder

processor

decoder



$AIFS_t \rightarrow t+6h$

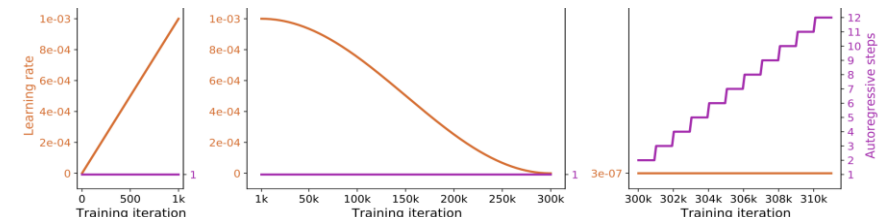
Lang et al. (2024)

Training Scheme

Step 1: pre-training phase, during which the model is given the task to forecast 6 h ahead

Step 2: model is trained to produce forecast up to 72 hr ahead

Step 3: model is fine-tuned on operational real-time analysis



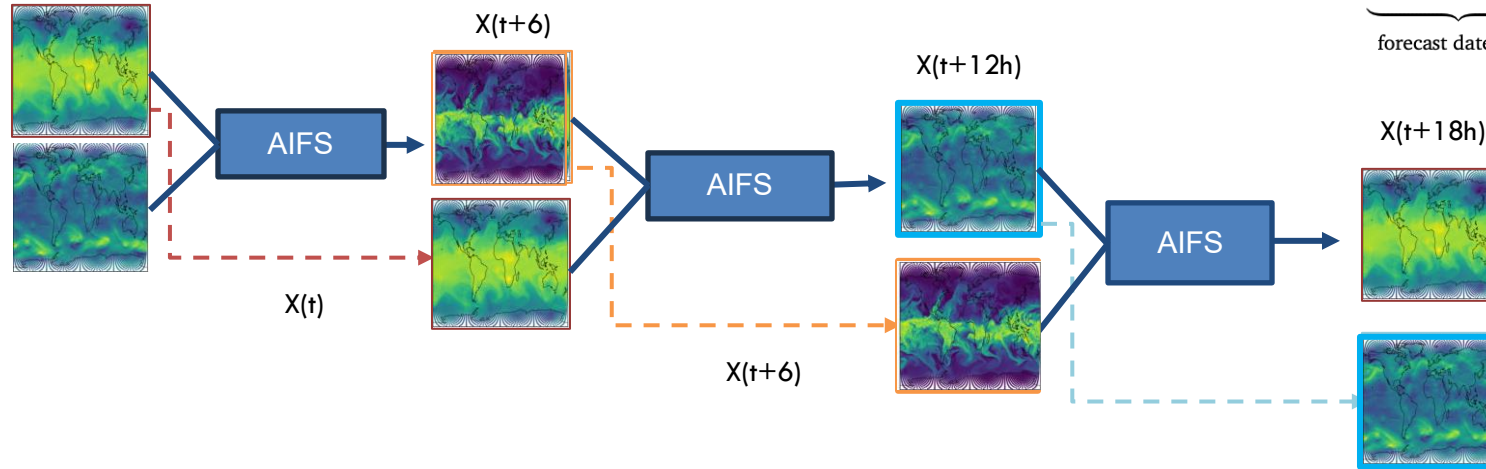
Scheme – Step 1

Scheme – Step 2 & 3

AIFS Single v1 - Training

TRAINING SCHEME – Step 2

Atmospheric state:
 $X(t), X(t-6h)$



WMSE - area-weighted mean squared error

$$\mathcal{L}_{\text{MSE}} = \underbrace{\frac{1}{|D_{\text{batch}}|}}_{\text{forecast date-time}} \sum_{d_0 \in D_{\text{batch}}} \underbrace{\frac{1}{T_{\text{train}}}}_{\text{lead time}} \sum_{\tau \in 1:T_{\text{train}}} \underbrace{\frac{1}{|G_{0.25^\circ}|}}_{\text{spatial location}} \sum_{i \in G_{0.25^\circ}} \underbrace{\sum_{j \in J}}_{\text{variable-level}} s_j w_j a_i \underbrace{(\hat{x}_{i,j}^{d_0+\tau} - x_{i,j}^{d_0+\tau})^2}_{\text{squared error}}$$

AR predictions
Up to rollout 12 (72 hours)



model is trained to produce forecast up to 72 hr ahead by aggregating the WMSE

ROLLOUT 1 - 6h

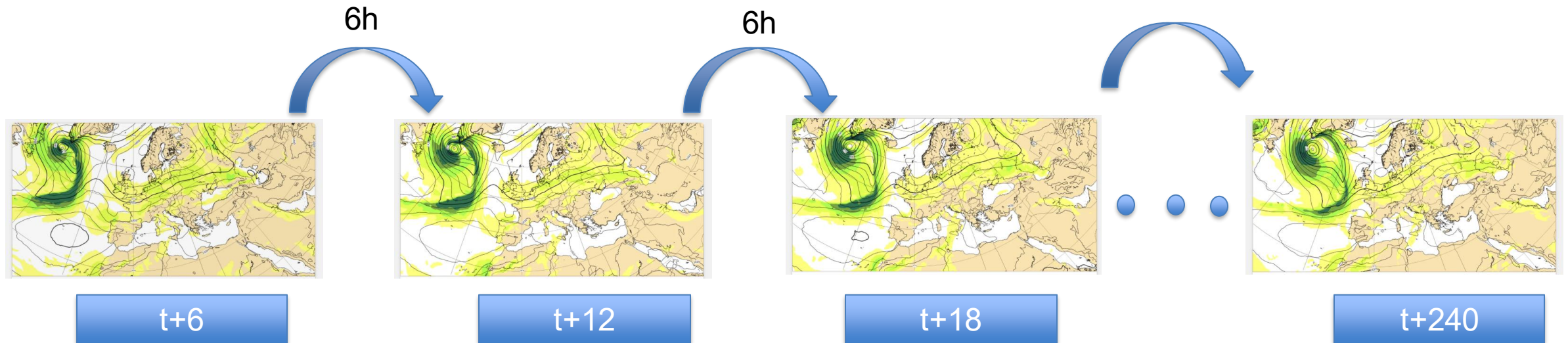
ROLLOUT 2 - 12hr

ROLLOUT 3 - 18hr

ROLLOUT 12- 72hr

AIFS Single v1 - Inference

Once the model is trained, we can generate predictions stepping e.g. 6h from analysis to analysis

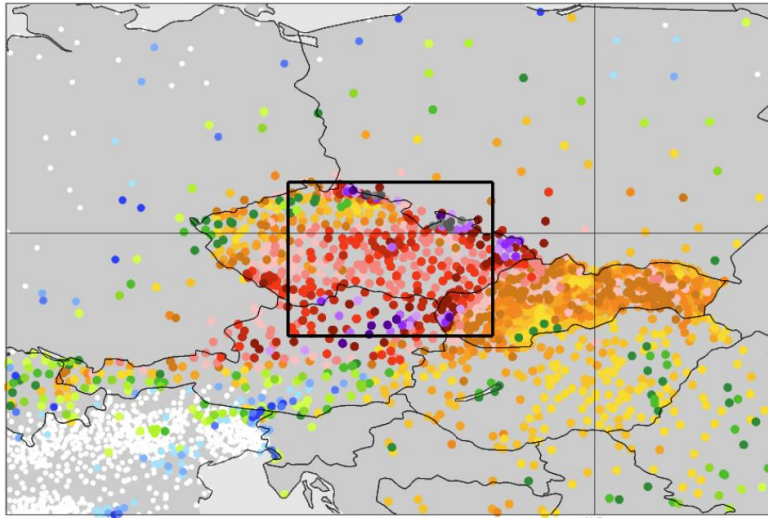


The forecast is then autoregressively stepping 6h into the future $x_n = f(x_{n-1}) \dots$

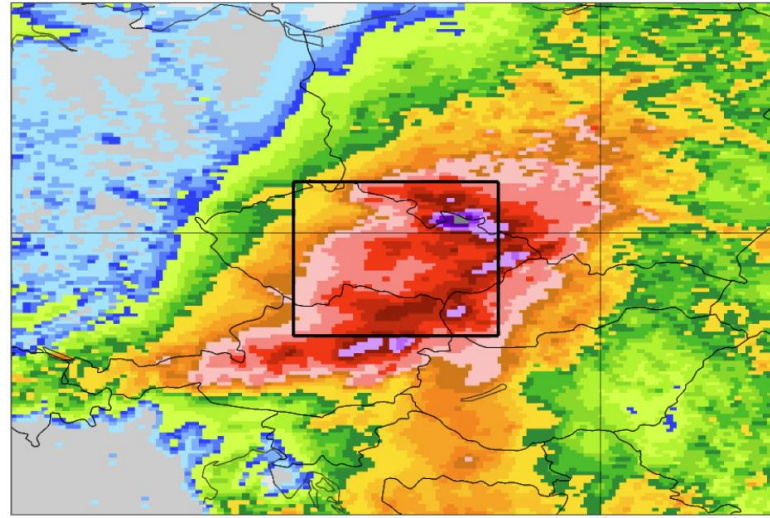
Extreme precipitation during Storm Boris in central Europe, September 2024

72-hour precipitation 13 September 00UTC – 16 September 00UTC

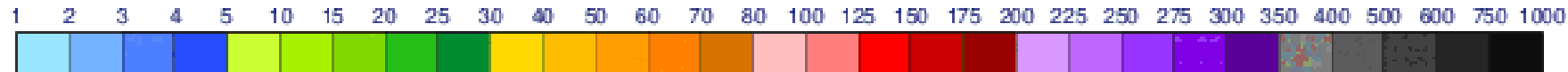
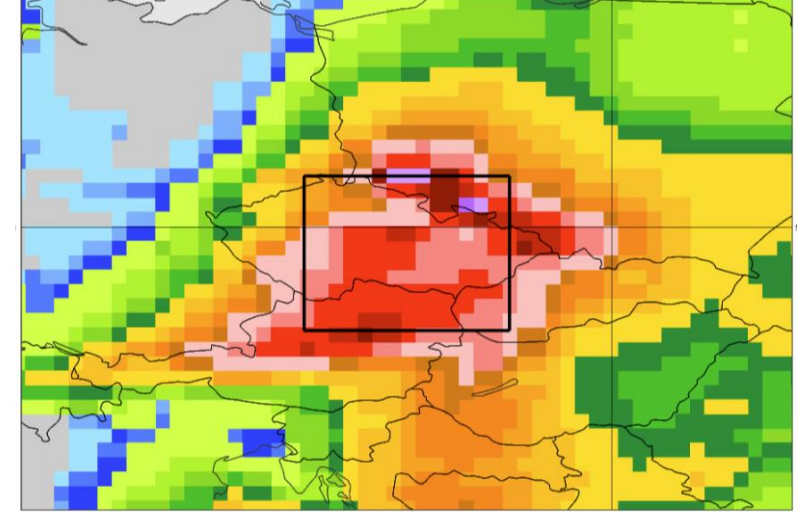
Observations



IFS control (48-120h)



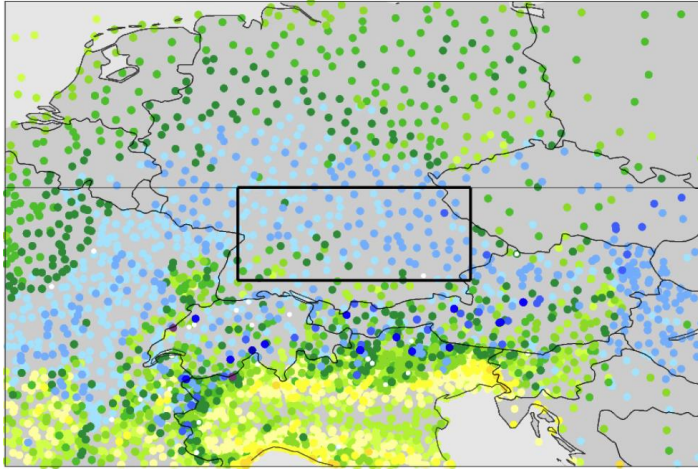
AIFSv1.0 (48-120h)



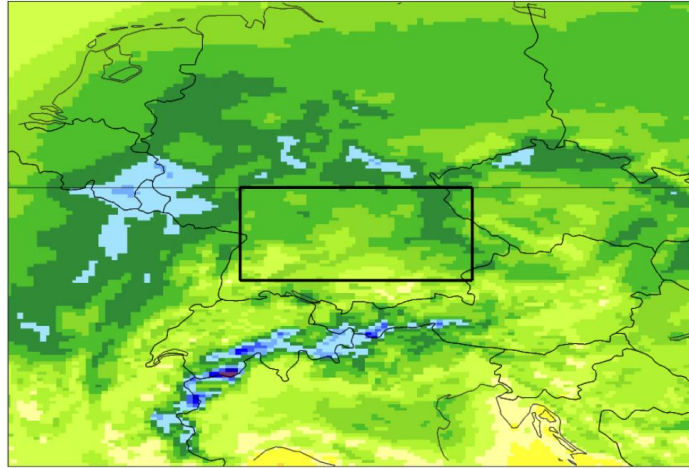
- AIFS predicted very extreme values for this region
- AIFS had a more consistent signal of the event over successive forecasts
- Smooth precipitation field from AIFS (do not capture local structures)

Cold Event in Europe in December 2024

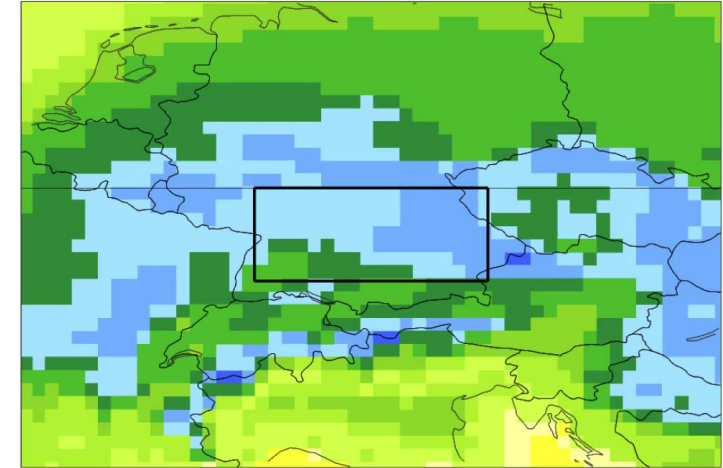
Observations



48-hour IFS Control



48-hour AIFS

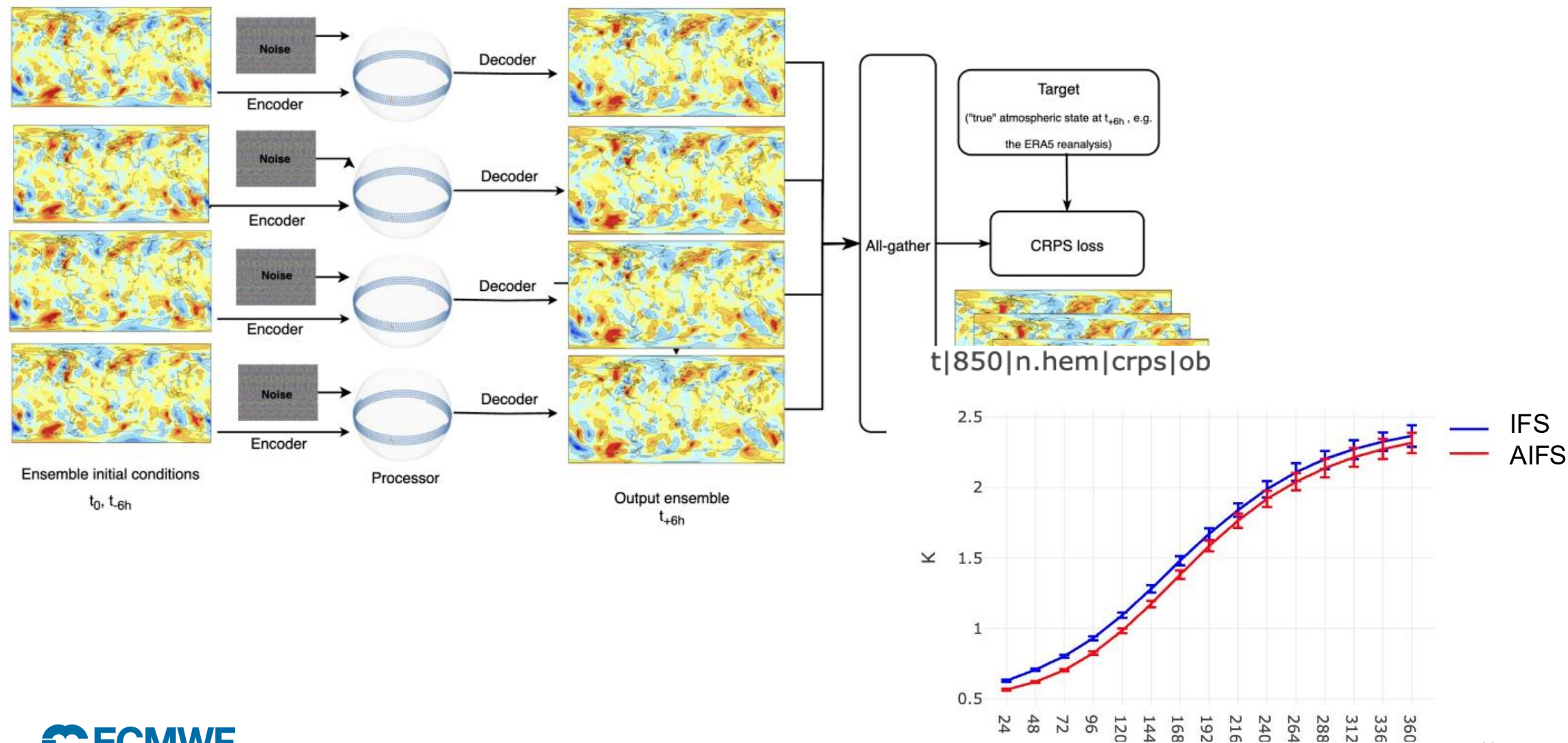


Karmouche et al. (2025):

AIFS blog: <https://www.ecmwf.int/en/about/media-centre/aifs-blog/2025/verifying-2-m-temperature-forecasts-wintertime-anticyclonic>

AIFS CRPS

Lang et al 2024b
Operational system from 1/7/25



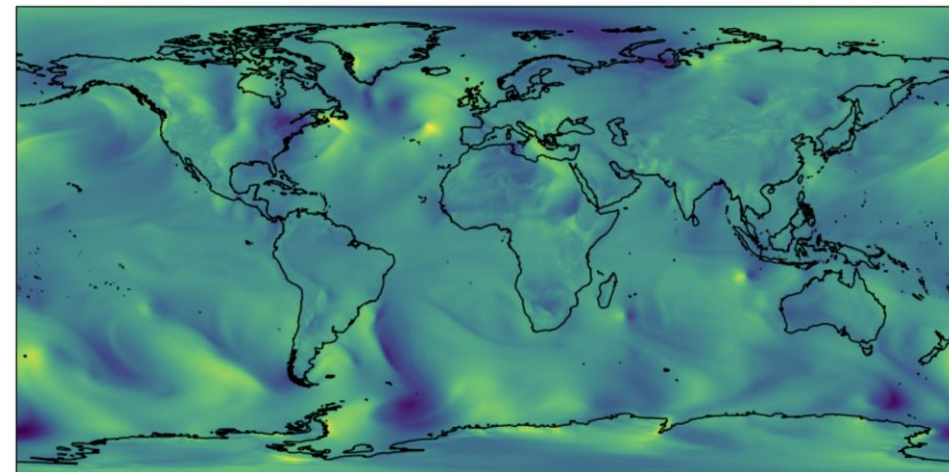
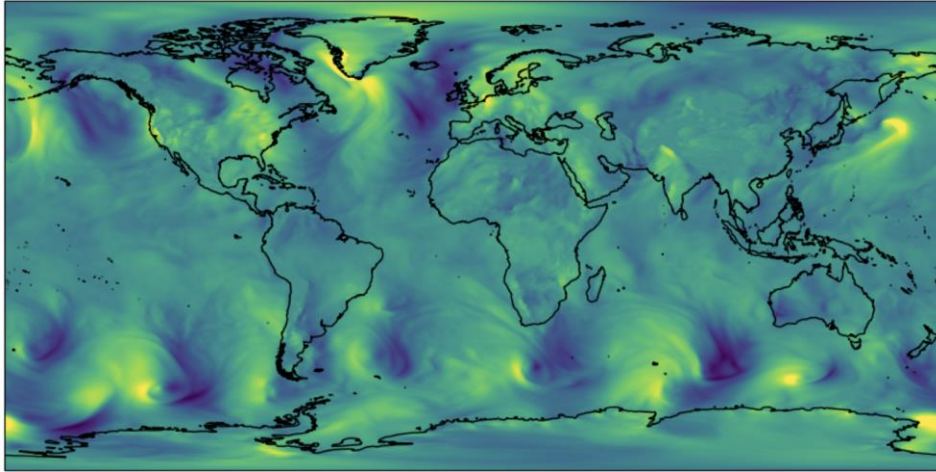
AIFS CRPS: CRPS training loss avoids smoothed forecasts

meridional wind at 850 hPa

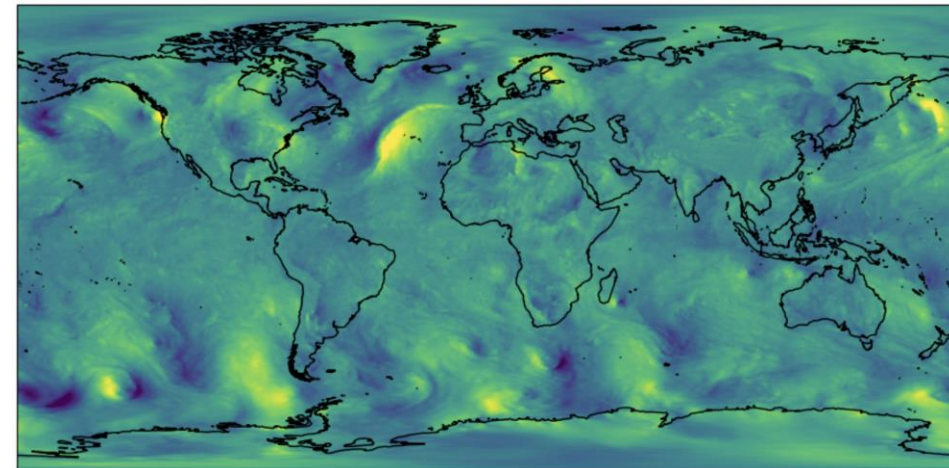
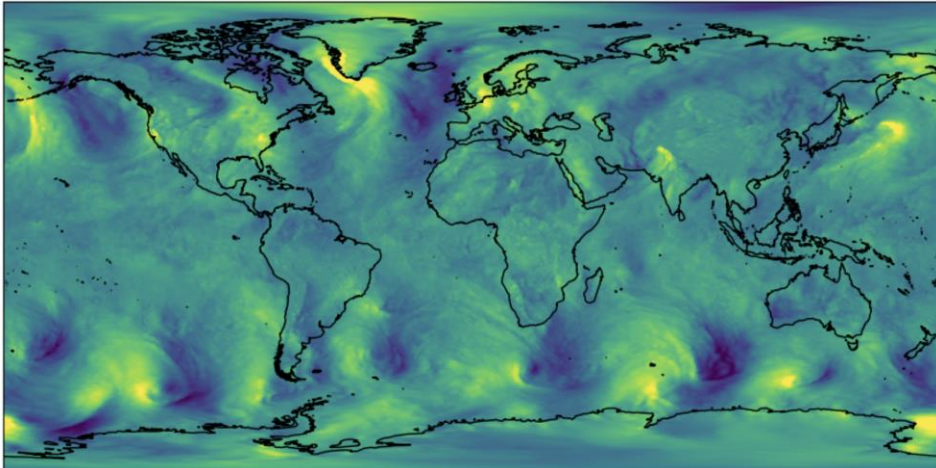
1-day forecast

10-day forecast

AIFS Single

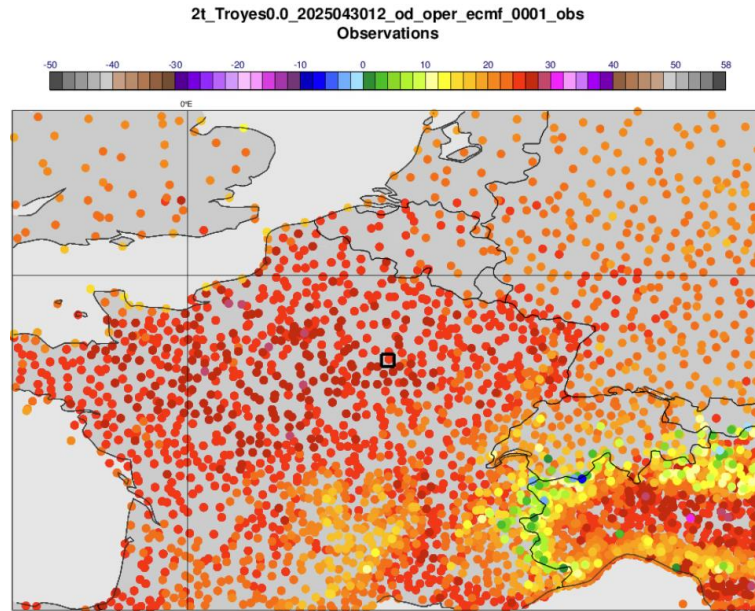


AIFS CRPS

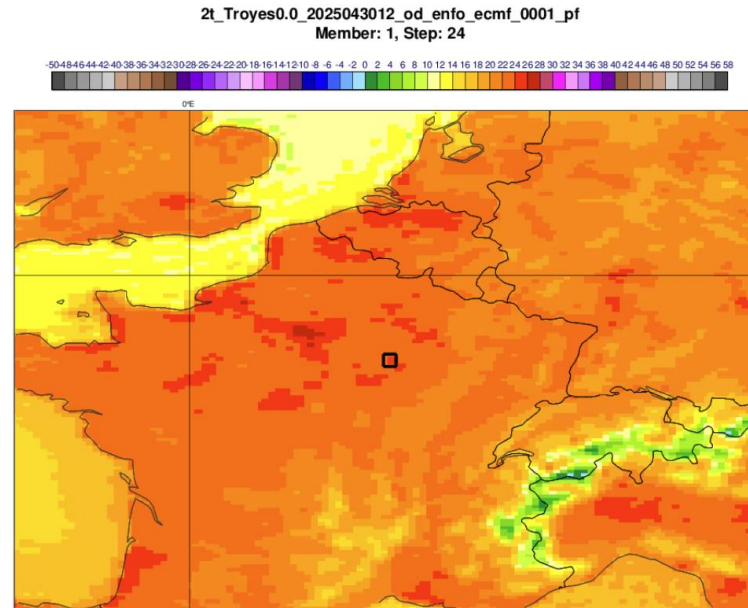


Early heatwave in April 2025: 2-metre temperature 30 April 12UTC

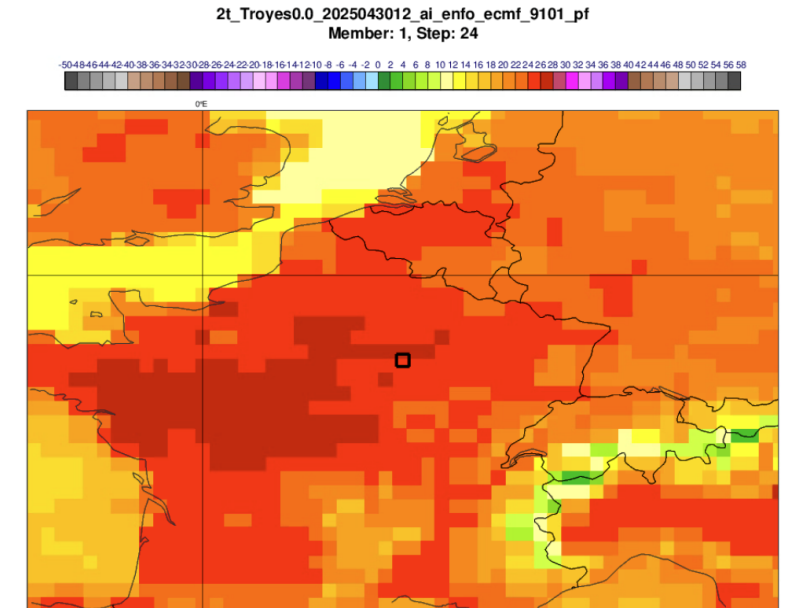
Observations



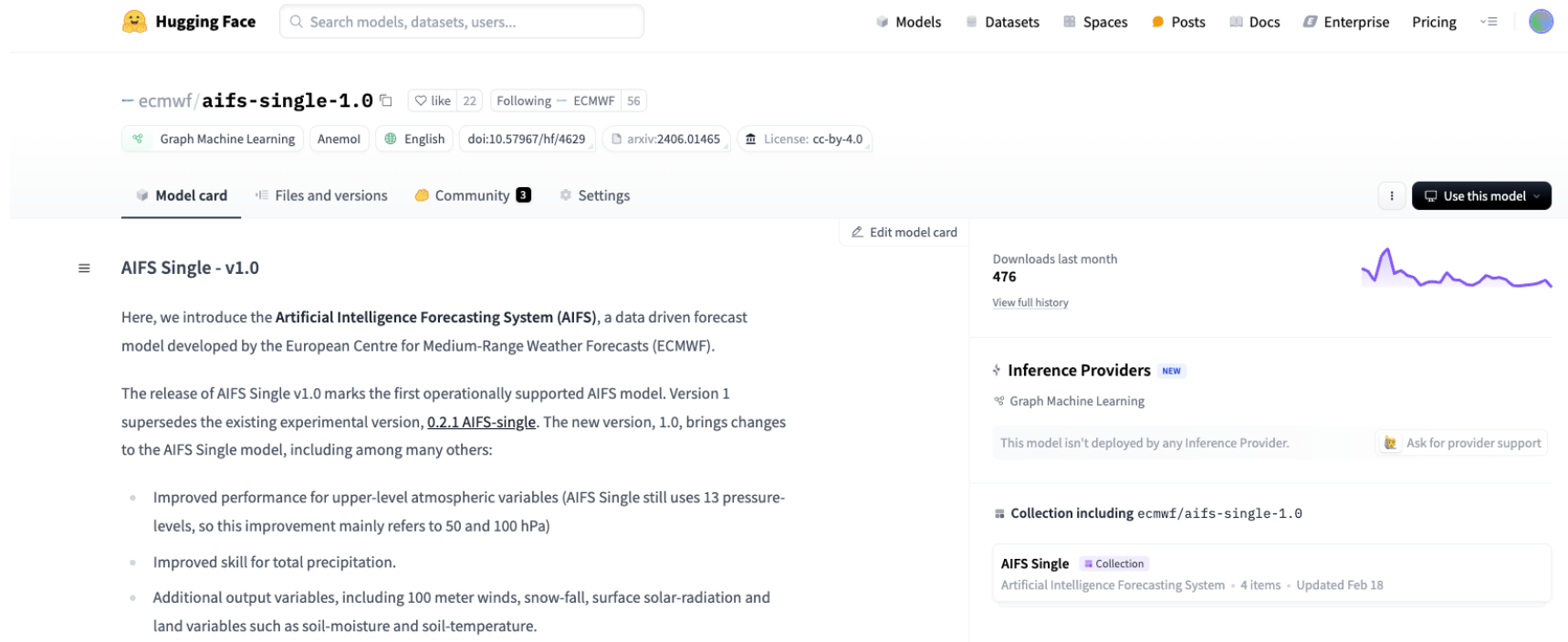
24h forecast from IFS-ENS Mem 1



24h forecast from AIFS-ENS Mem 1



Running AIFS yourself interactively



The screenshot displays the Hugging Face interface for the model `ecmwf/aifs-single-1.0`. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Posts, Docs, Enterprise, and Pricing. The model page header shows the model name, a like button (22 likes), a following button (56 following), and tags for Graph Machine Learning, Anemol, English, and a license of cc-by-4.0. The main content area is divided into two columns. The left column contains the model card, which includes a description of the Artificial Intelligence Forecasting System (AIFS) model, its version history, and a list of features. The right column shows the download history, a line graph of downloads over time, and a section for inference providers. The model is currently not deployed by any inference provider.

Model card | Files and versions | Community | Settings

AIFS Single - v1.0

Here, we introduce the **Artificial Intelligence Forecasting System (AIFS)**, a data driven forecast model developed by the European Centre for Medium-Range Weather Forecasts (ECMWF).

The release of AIFS Single v1.0 marks the first operationally supported AIFS model. Version 1 supersedes the existing experimental version, [0.2.1 AIFS-single](#). The new version, 1.0, brings changes to the AIFS Single model, including among many others:

- Improved performance for upper-level atmospheric variables (AIFS Single still uses 13 pressure-levels, so this improvement mainly refers to 50 and 100 hPa)
- Improved skill for total precipitation.
- Additional output variables, including 100 meter winds, snow-fall, surface solar-radiation and land variables such as soil-moisture and soil-temperature.

Inference Providers NEW

Graph Machine Learning

This model isn't deployed by any Inference Provider. [Ask for provider support](#)

Collection including ecmwf/aifs-single-1.0

AIFS Single Collection

Artificial Intelligence Forecasting System • 4 items • Updated Feb 18



AIFS Single v1.1



AIFS Ens v1.0

- AIFS Single v1.0 and AIFS Ens v1.0 on Hugging Face – an open platform for sharing the models.
- Includes interactive notebook on how to run from ECMWF open data, can be adapted for other sources.

3. Load the Model and Run the Forecast

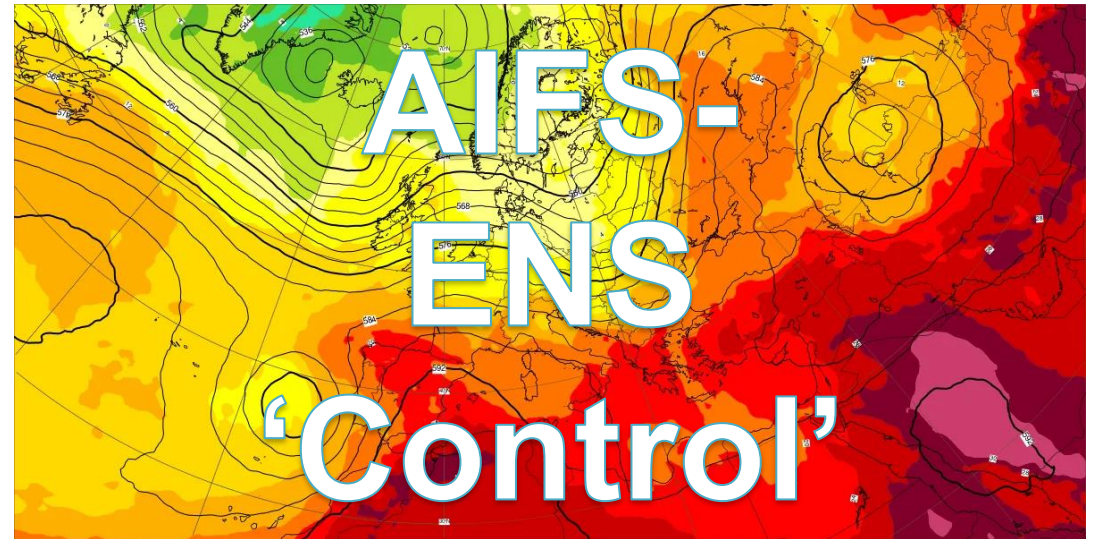
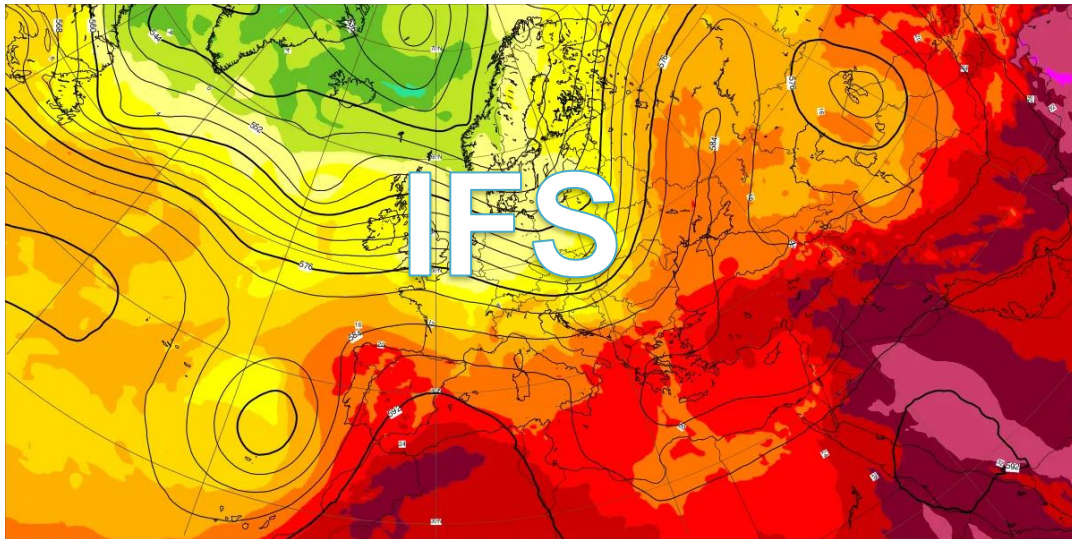
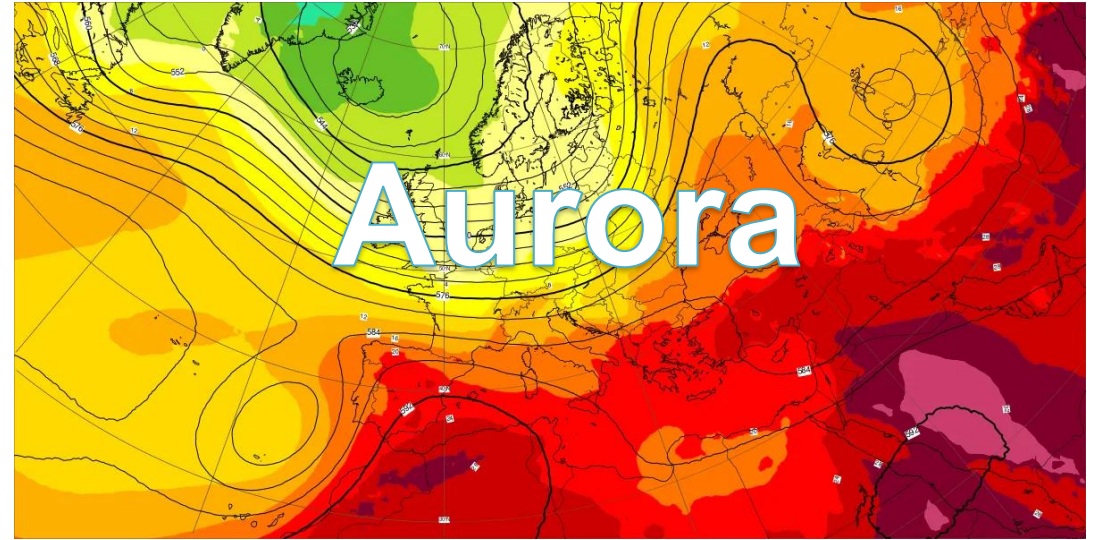
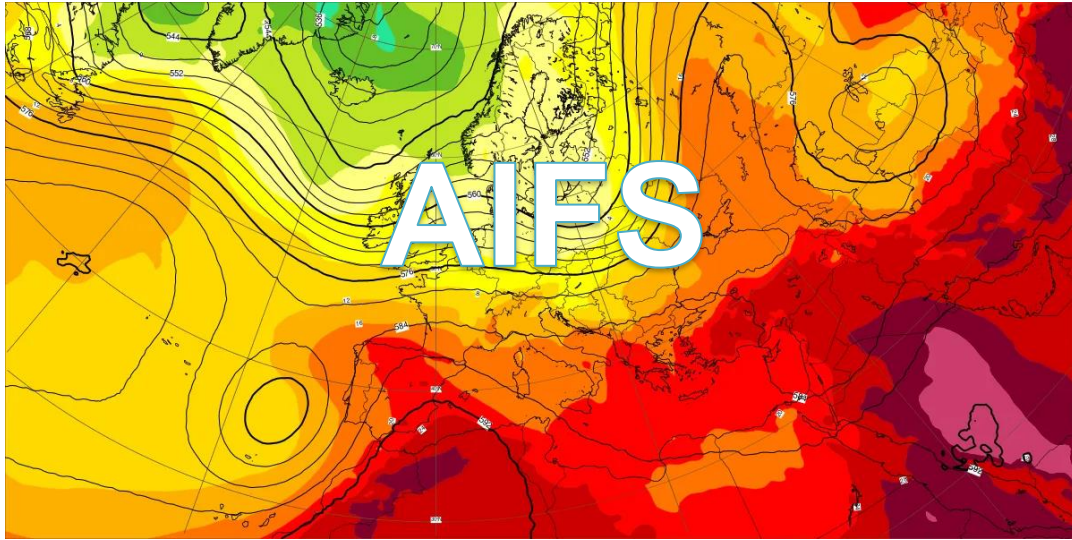
Download the Model's Checkpoint from Hugging Face & create a Runner

```
In [13]: checkpoint = {"huggingface": "ecmwf/aifs-single-1.0"}
```

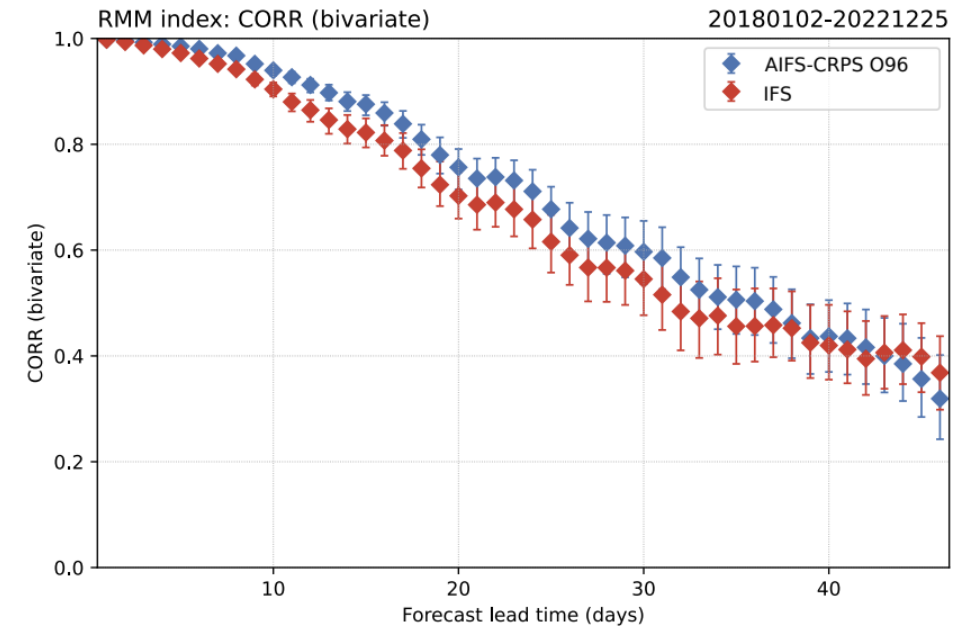
T+156

Basetime: Tue 03 00 UTC

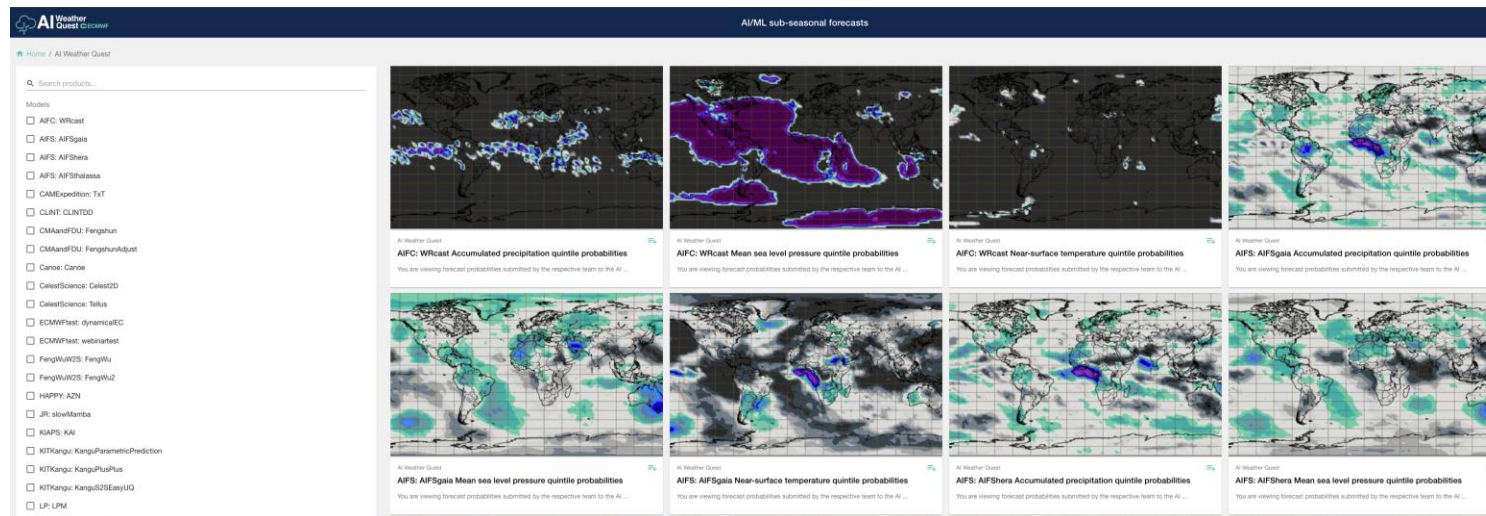
Validtime: Mon 09 12 UTC



AIFS CRPS already has clear skill on sub-seasonal
(*Lang et al. 2024b*)

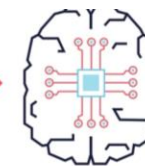


AI Weather Quest taking place currently, challenging participants to produce skilful sub-seasonal weather forecasts with AI

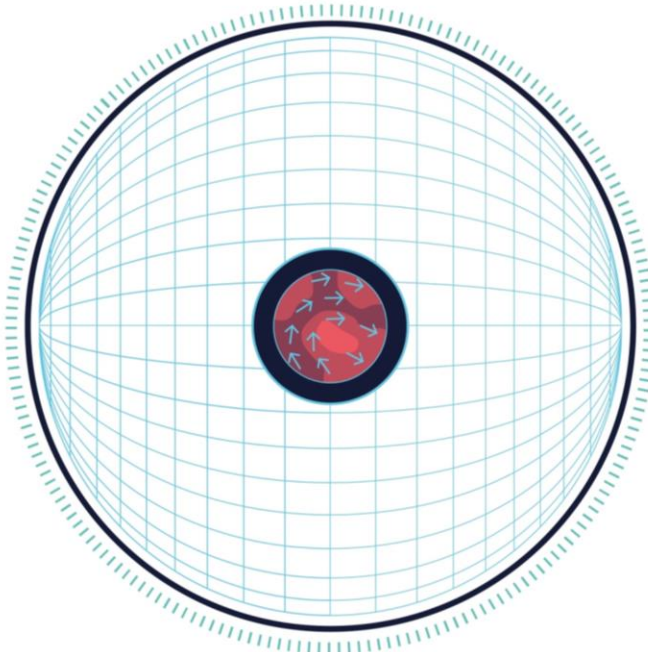




EXPANDING TOWARDS AN EARTH-SYSTEM AI MODEL WITH DESTINE



ATMOSPHERIC COMPONENT



FORECASTING THE WEATHER
(ECMWF'S AIFS)



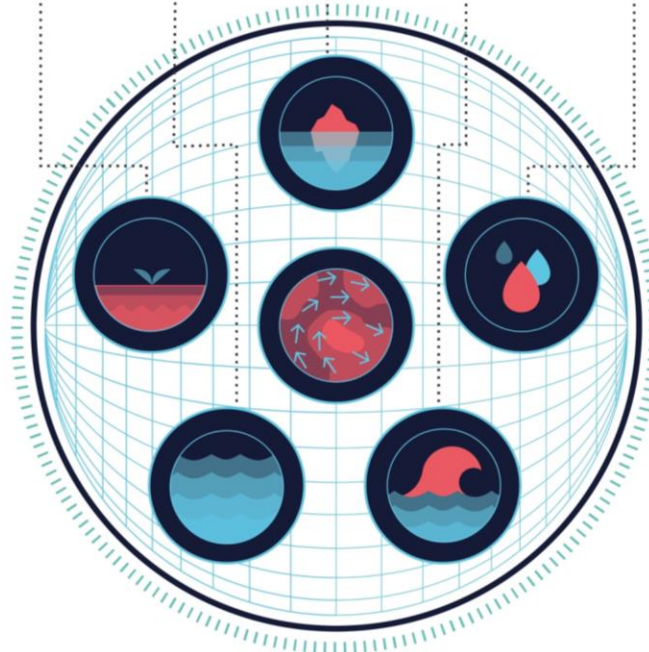
LAND

OCEAN

SEA-ICE

WAVE

HYDROLOGY

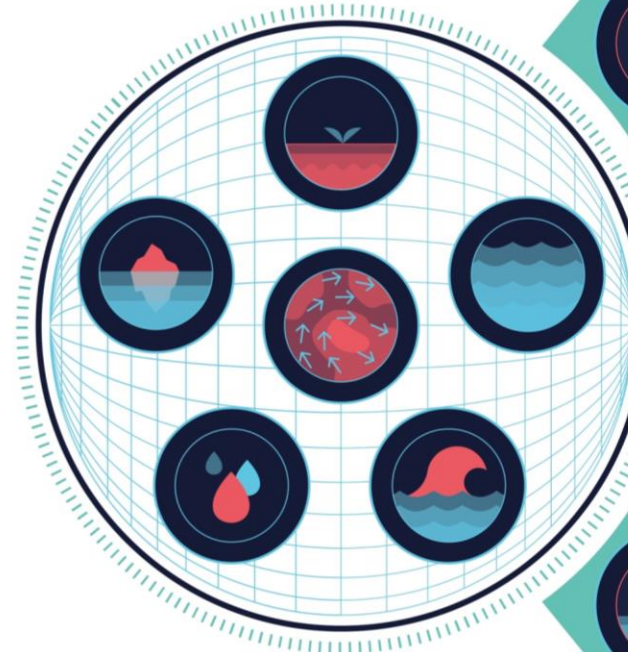


FOR WEATHER
EXTREMES

FOR CLIMATE
PROJECTIONS

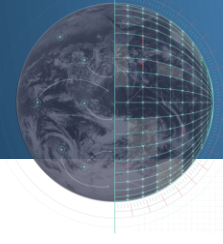


IMPACT SECTORS

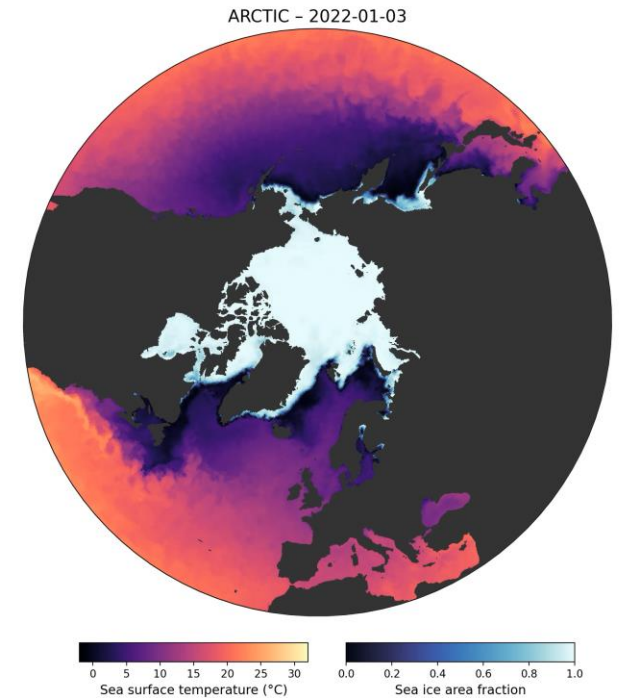
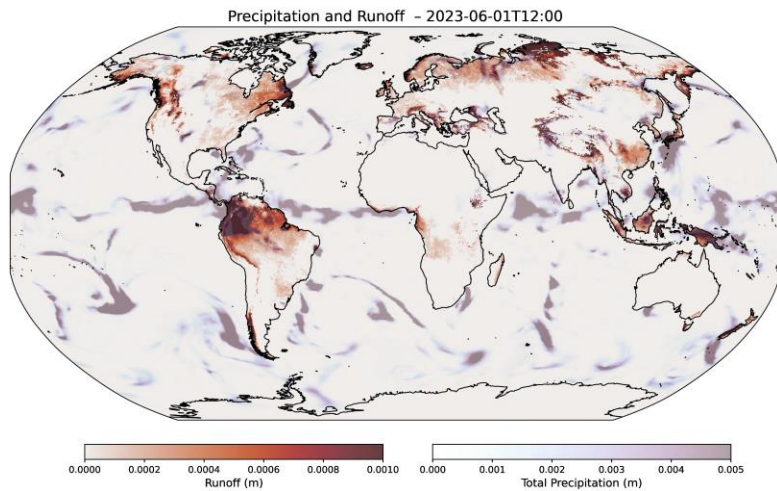
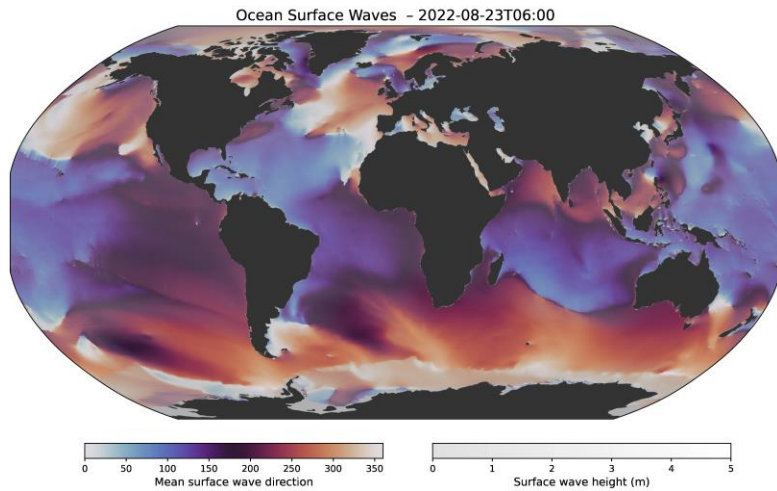


DEMONSTRATORS FOR
IMPACT SECTORS





Status: First prototypes across the Earth System



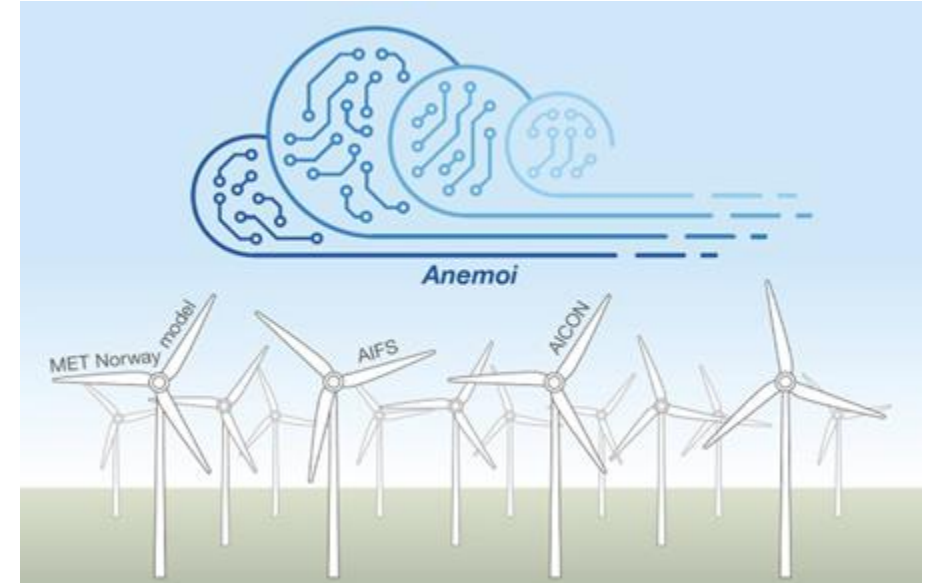
Anemoi: the tool behind it all

What is Anemoi?

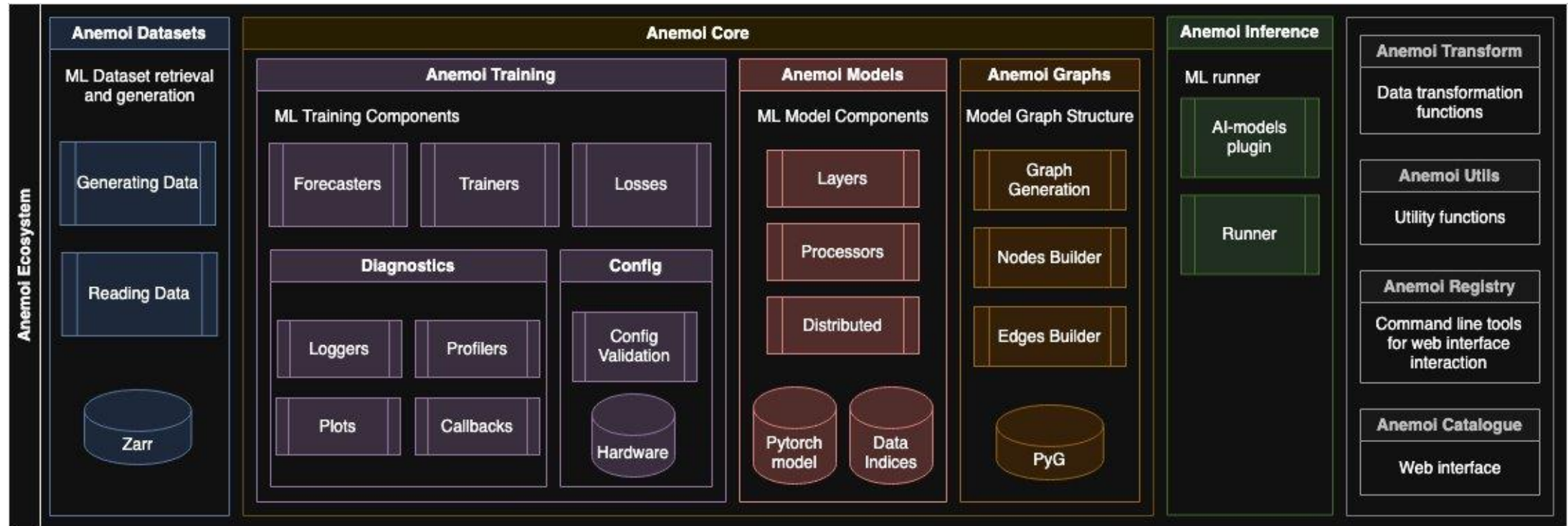


What is Anemoi?

- An *open-source framework* that provides a complete toolkit to develop data-driven weather models – from data preparation through to inference
- Empowering *real-time running* and operationalisation of data-driven models at meteorological centres
- A *highly modular framework* organised into different Python packages



What is Anemoi?



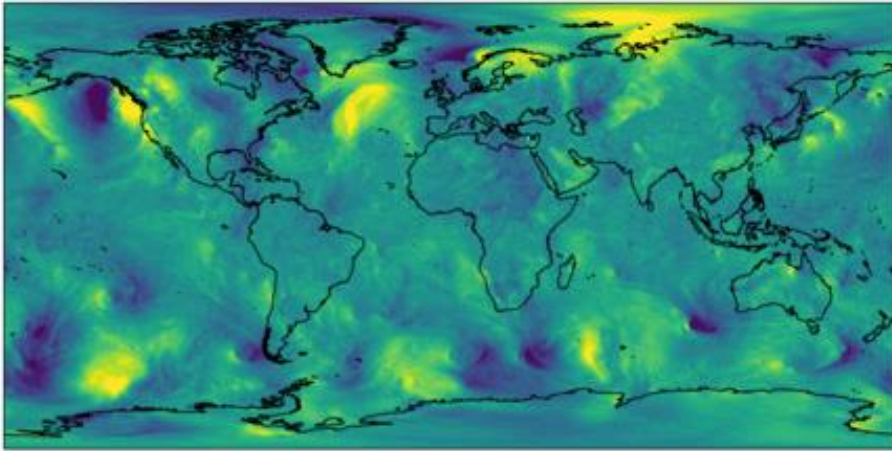
Who is Anemoi?

Pooling of resources without resulting in a single forecasting model.



Open-source development driven by European Meteorological Centers and ECMWF

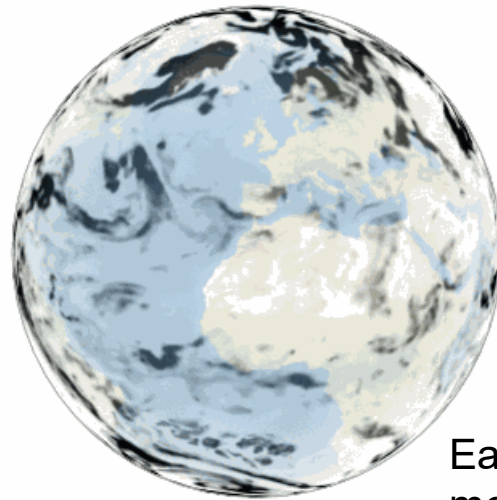
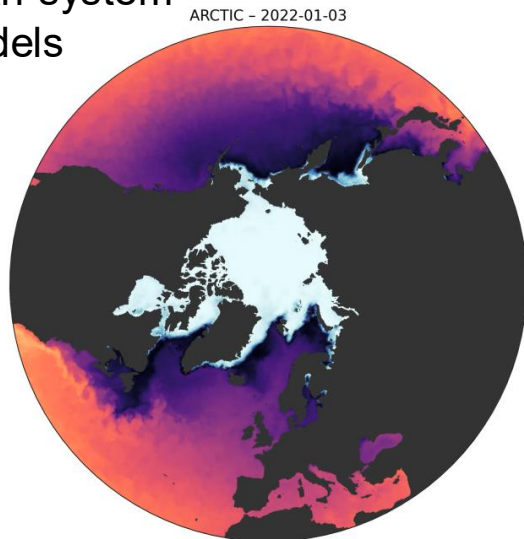
What Anemoui enables?



Training deterministic & ensemble models

Lang et al. 2024a & b

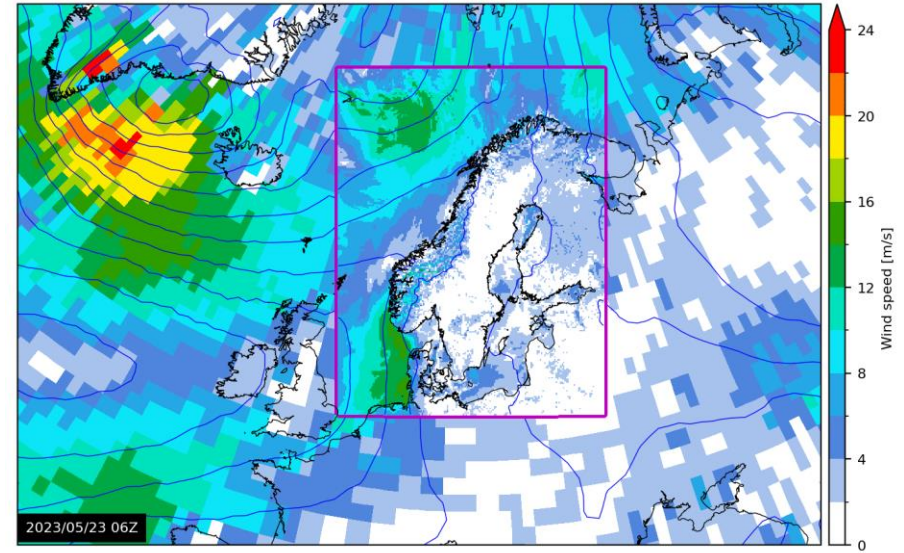
Building coupled earth-system models



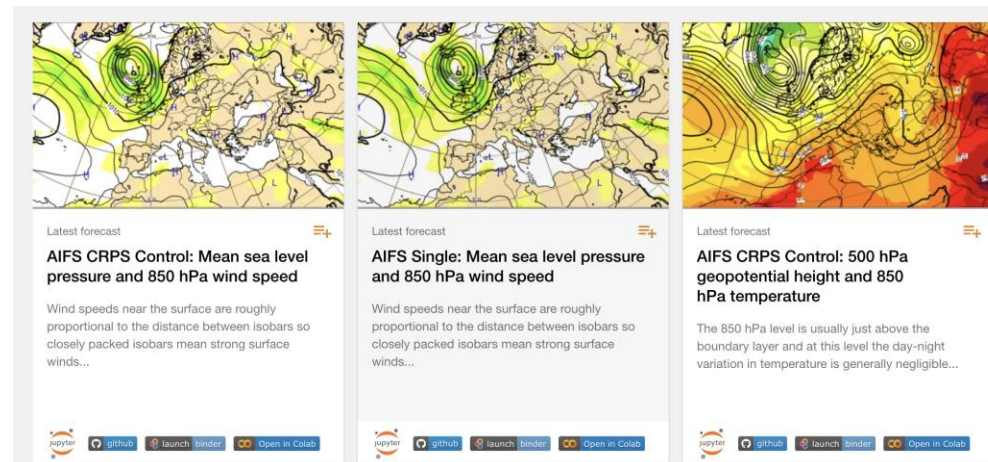
Easy exploration of more atmospheric fields

Regional high-resolution modelling:

Nipen et al. 2024



Real-time running at operation centres
Anemoui is powering ECMWF's operationalisation of AIFS in 2025.

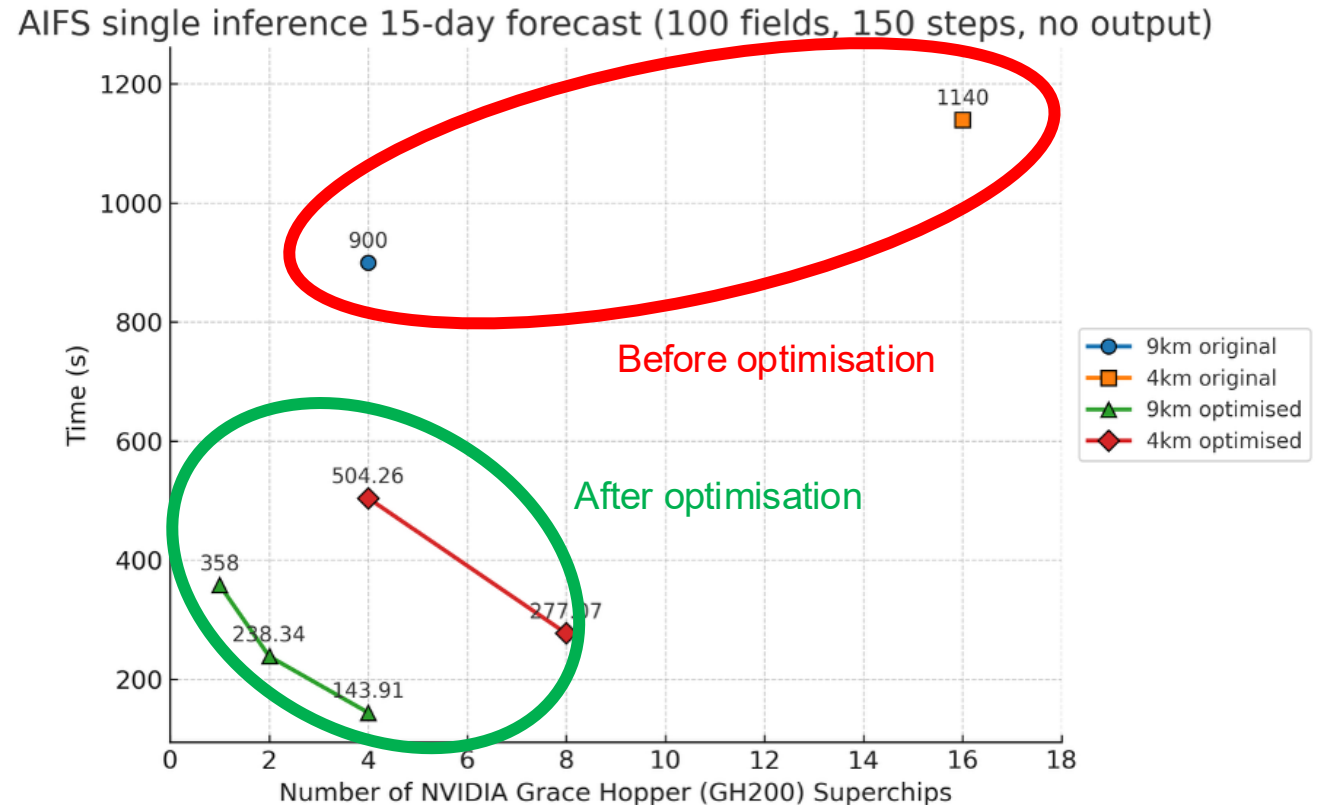


Anemoi – efficient training and inference

*More on optimising Anemoi
& AIFS from Cathal O'Brien
and Jan Polster on
Wednesday*

Key value of collaboration – community
can work/benefit together from
algorithmic performance improvements.

Example: improvements in time & GPU
utilisation for experimental high-
resolution inference.

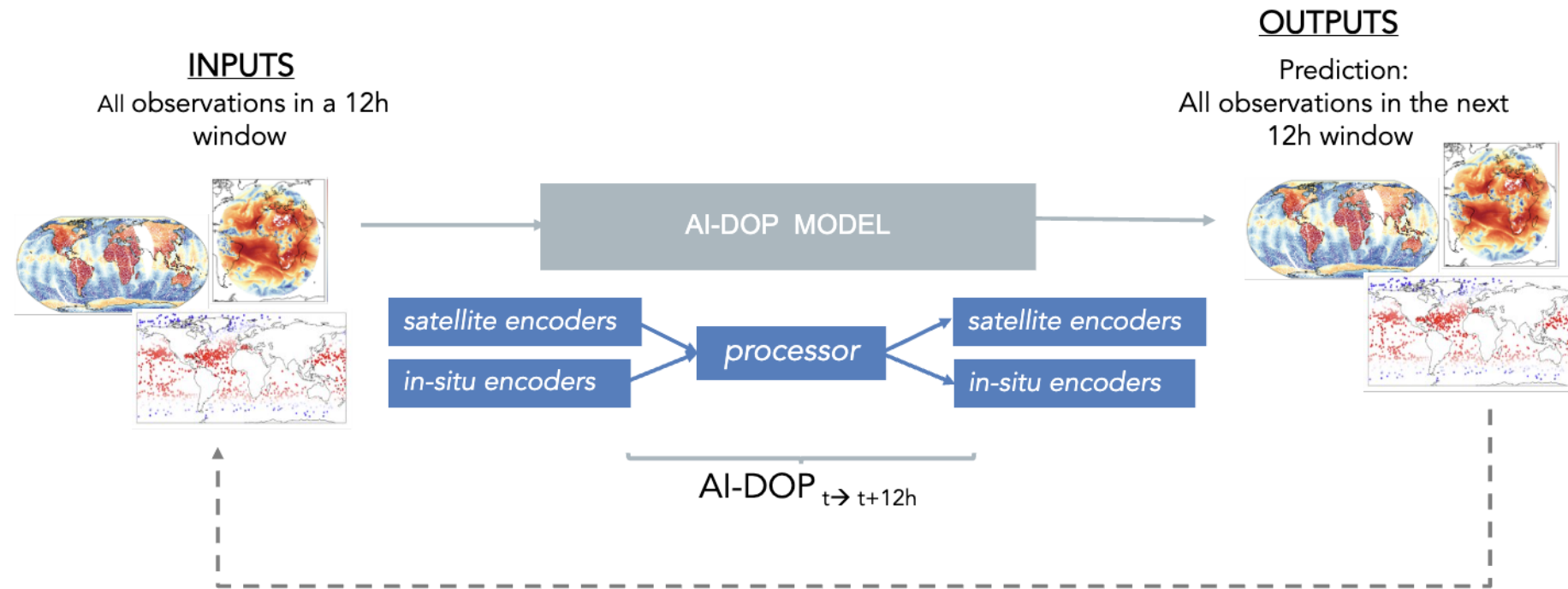


Learning directly from observations

AI-DOP: Emulating the entire chain using only observations

Alexe et al. 2024

- Use historical observations to train a Neural Network to forecast future observations (don't need analysis)
- Include all available observations of the full Earth system (atmosphere, ocean, land) simultaneously

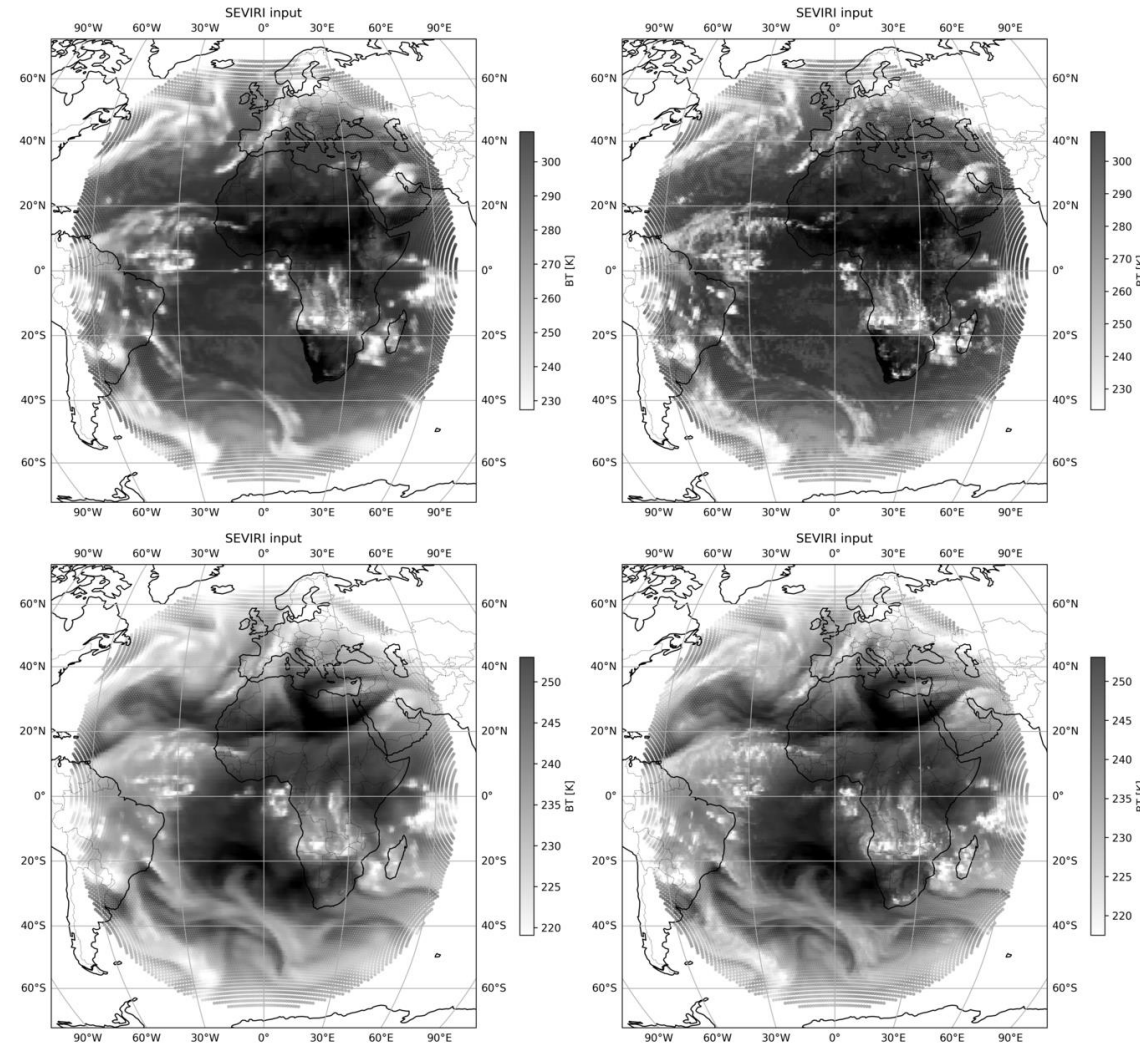


AI-DOP: Emulating the entire chain using only observations

Alexe et al. 2024

SEVIRI prediction

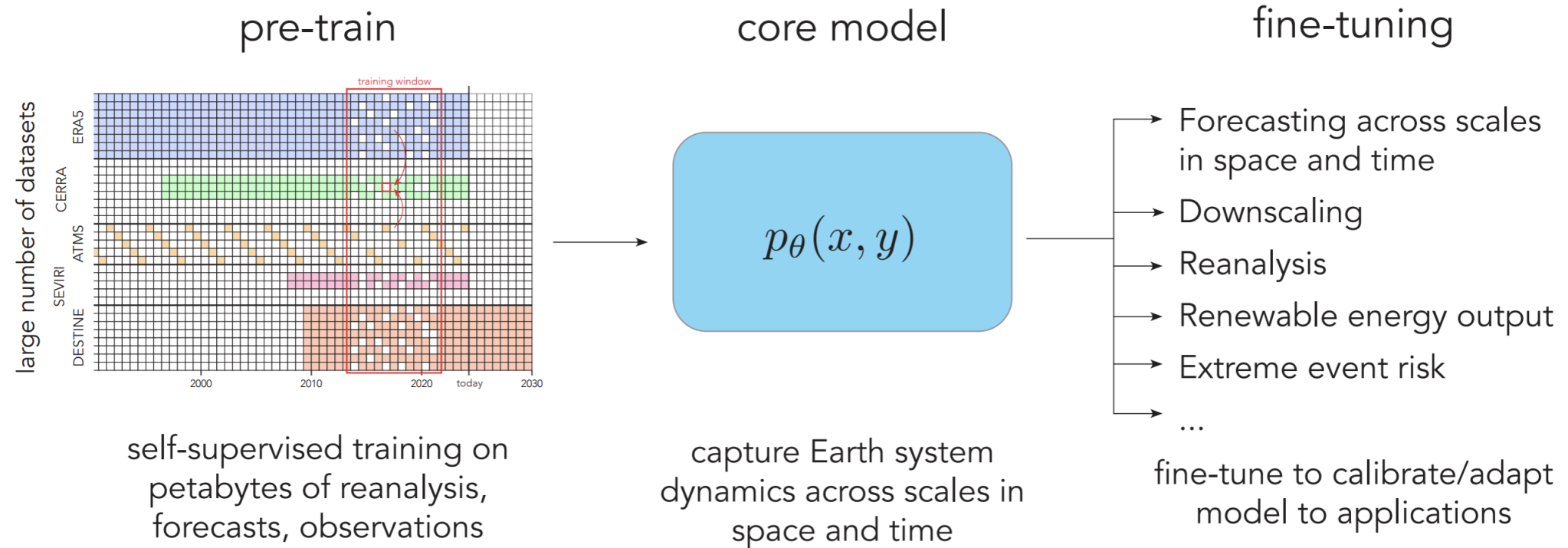
SEVIRI target



WeatherGenerator



A pan-European initiative to build a single open-source foundational model for weather and climate



Summary

- Data-driven weather forecasting is a fast-evolving and exciting landscape with models already operational after a couple of years of development
- And there's more to come:
 - Full Earth System models
 - Direct Observation Prediction models
 - Foundational models
 -

Model Checkpoints



AIFS Single v1.1



AIFS Ens v1.0

Code



Anemoi code tour



Anemoi GitHub Repo



WeatherGenerator
GitHub Repo

Papers



AIFS



AIFS Ens



Graph-DOP

AIFS ENS Forecast

(7 days ahead)

AIFS ENS Meteogram

Dublin - Leinster - Ireland 53.26°N 6.3°W (ENS land point) 3 m

AIFS ENS Distribution Monday 6 October 2025 00 UTC

